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Multi-User Shore Base Facilities: Redefining Marine Logistics for the Blue Economy

Carmelo (sive Karl) Bartolo, Dpty. CEO MedservRegis PLC, Port of Marsaxlokk, Birzebugga, Malta

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Abstract

This paper presents a transformative approach to marine logistics for offshore oil and gas projects, proposing the implementation of multi-user shore base facilities. These shared logistics hubs are designed to serve both National Energy Companies (NECs) and International Energy Companies (IECs) across all phases of offshore development: exploratory, development, and production. By transitioning from siloed, company-specific operations to centralized, collaborative logistics models, the approach aims to reduce costs, improve efficiency, and minimize environmental impact. This model aligns with the broader goals of the Blue Economy by promoting sustainability, digitalization, and the optimized use of infrastructure and resources. The paper outlines the methods, procedures, and results associated with this logistics model, and argues for a fundamental mindset shift in the industry to embrace shared infrastructure and digital transformation.

Introduction

Marine logistics underpin the entire offshore oil and gas industry, functioning as the operational lifeline that connects drilling rigs and production platforms to onshore supply networks. Yet, despite their criticality, these logistics systems often suffer from inefficiencies rooted in fragmentation, redundancy, and a lack of coordination. Traditional shore base facilities are developed and operated independently by each oil company, leading to duplicated infrastructure, inconsistent operational standards, and unnecessary environmental impact and financial costs.

At the same time, the global energy sector is under growing pressure to transition toward cleaner, more sustainable operations. Governments, regulators, investors, and society at large are demanding reduced emissions, increased transparency, and efficient resource use. The concept of the **Blue Economy**—centered around the sustainable use of ocean resources—has emerged as a powerful framework to align economic activity with environmental stewardship. For offshore oil and gas operators, this means rethinking how infrastructure is developed, used, and shared to minimize environmental footprints while improving efficiency.

Throughout the years of working in the industry I have seen many at times various oil companies operating in silos, with minimal sharing of shore base facilities and Platform Supply Vessels (PSVs), and in

my opinion the main reason for resisting a change in mindset is resistance to change due to fear of failure in the Logistics Supply Chain.

This paper introduces **multi-user shore base facilities** as a transformative marine logistics model that meets the operational needs of both **National Energy Companies (NECs)** and **International Energy Companies (IECs)** while aligning with the goals of the Blue Economy. By transitioning from siloed to centralized logistics hubs that serve multiple stakeholders, this model promises significant cost savings, improved operational efficiency, and a measurable reduction in emissions and waste. It promotes digital integration, shared infrastructure, standardized Quality, Health, Safety, Security and Environment QHSSE procedures, and a pay-per-use structure that collectively redefine how offshore logistics are managed—offering a blueprint for a more agile and sustainable offshore energy sector.

Identifying the Problem & Studies on Shared Logistics

Existing practices by National Energy Companies (NECs) and International Energy Companies (IECs) is establishing independent logistics networks in order to maintain control, ensure security, and protect proprietary operations. However, this approach leads to duplicated facilities, underused assets, and increased emissions.

Studies have shown that redundant vessel trips and fragmented shore base operations contribute significantly to operational costs and environmental degradation. Additionally, siloed logistics result in inconsistent service quality and resilience, particularly during supply chain disruptions, such as those witnessed during the COVID-19 pandemic. These inefficiencies are at odds with the objectives of the Blue Economy, which seeks to maximize sustainable benefits from marine resources.

A study titled '*Exploring More Sustainable Offshore Logistics Scenarios Using Shared Resources: A Multi-Stakeholder Perspective*' by Idriss El-Thalji in 2024 had quantified the benefits and impacts of applying the shared logistics concept from several points of view (asset operator, vessel contractors, and authority). The results of his study were the following:

KPIs from Asset Operator Perspective. The oil and gas operator is mainly interested in getting the requested oil and gas equipment as fast as possible to avoid any associated production downtime. Thus, the supply lead time and vessel availability are critical to satisfy their needs. Vessel availability is on-demand availability which means the vessel will be ready for shipping whenever the operator needs it. However, the delivery frequency, i.e., the time between deliveries, is a better measure to show the operational availability for on-demand assets like vessels; for example, if a vessel is delivering every day, that means it is available on a daily basis. The enhancement in the delivery frequency is extremely significant to operators and gives them the impression of higher vessel availability. Furthermore, the delivery frequency has enhanced the number of deliveries. In fact, the operator did not expect a reduction in the number of deliveries as it follows the demand; however, the shared logistics concept reinforced the batching action more where several oil and gas equipment and consumables were delivered in one delivery.

KPIs from Vessel Contractor Perspective. The vessel contractors are mainly interested in satisfying their customers (asset operators) and enhancing vessel utilization, cutting down the operating costs and carbon emissions, thereby generating more revenue. The result of the case study by Idriss El-Thalji is the reduction in the total deliveries or journeys for each vessel has led to a reduction in the utilization rate for Vessels. However, in a scenario where each vessel is contracted for less period the utilization rates during the period contracted have increased and available time to contract additional customers is offered. Improving the utilization rate increasing the availability to generate additional revenue. It is obvious that the scheduled shared logistics scenario has an advantage for the vessel contractor over the on-demand shared scenario as it provides a higher utilization rate and revenue-generating periods.

KPIs from Authority Perspective. From an authority point of view, shared logistics initiatives will improve the volume of maritime traffic (sea crowding) and use of port facilities, reduce congestion, and cut down carbon emissions.

In summary, there is around one-third carbon emission reduction once the shared logistics concept is utilized. (El-Thalji, 2024; El-Thalji, 2024)

MedservRegis Case Study

While the previous study focused on the client and its stakeholders, MedservRegis conducted an independent analysis of a shore base supporting two distinct International Energy Companies (IECs). This study aimed to understand existing work practices and highlight the benefits of shared logistics—specifically from the perspective of the shore base operations.

The evaluation covered several key operational areas: Quayside Processes, Bunkering Operations, Warehousing, Management of Oil Country Tubular Goods (OCTG), Vessel Scheduling, Equipment Management, and Vessel Operations. It identified critical issues and potential improvements to enhance the effectiveness of shared logistics.

Quayside Operations

Issues identified included vessel clustering at berthing areas, excessive loose cargo on the quayside, and a lack of recorded turnaround times for loading and discharge. Offshore teams were found to operate in silos, with minimal planning or look-ahead capability, hindering demand sharing. Key recommendations included:

- Increase the use of Cargo Carrying Units (CCUs) to consolidate cargo and reduce loose items, improving operational efficiency.
- Implement systematic recording of all loading and unloading times.
- Enhance scheduling through a centralized digital system that enables real-time, two-way communication across marine operations.
- Establish a 14-day look-ahead planning process for all user groups, managed by the logistics department.

KPIs

- Average vessel turnaround time (hrs per vessel at quayside)
- Quayside throughput (tons or CCUs per day/week)
- Percentage reduction in loose cargo handling (loose cargo → CCUs)

Bunkering Operations

Bunkering was found to be inefficient due to the reliance on dedicated vessels and fuel burn while vessels awaited quayside operations. Additionally, there was limited fuel monitoring and minimal GPS utilization to verify location and speed. Recommendations included:

- Optimize the current fleet for multi-purpose use.
- Implement digital fuel monitoring systems to track and reduce consumption.
- Engage vessel owners in performance discussions, focusing on fuel efficiency.

KPIs

- Fuel consumption per voyage (m³ or liters per trip)
- Fuel consumption per ton of cargo moved
- Percentage reduction in idle fuel burns (during waiting/queuing at quayside)
- CO₂ emissions reduction (tons of CO₂ avoided through fuel savings & vessel optimization)

Warehousing

Warehouse facilities were dispersed across the base, lacking consolidation and co-location, which negatively affected efficiency. The Warehouse Management System (WMS) required assessment due to the prevalence of scrap and slow-moving inventory. Suggested improvements included:

- Consolidate and cluster warehouse facilities throughout the marine base.
- Lease excess warehouse space to third-party users.
- Introduce a modern WMS to support the management of diverse inventory types and drive transformation.

KPIs

- Percentage reduction in warehouse footprint used (m² freed through consolidation & leasing out)
- Inventory turnover ratio (usage vs stock levels)

Oil Country Tubular Goods (OCTG)

Significant quantities of OCTG materials occupied open yard areas, with limited visibility into their condition (e.g., usable, damaged, or scrap). A key recommendation was the implementation of Vendor-Managed Inventory (VMI), which would include scrap disposal as part of lean operational practices, thereby freeing up valuable yard space and improve the Operations working capital cycle by turning inventory to cash.

KPIs

- Percentage scrap material reduction (OCTG & other slow-moving stock)
- Value of working capital freed (USD from disposing/recycling scrap)

Vessel Scheduling

Scheduling practices were not aligned with cargo needs. Voyages were conducted without consideration of cost or payload, and user demand data was not shared in advance with management. This lack of coordination resulted in silos and inefficiencies. Recommendations included:

- Implement a 14-day look-ahead plan and hold regular review meetings, introducing cut-off times for cargo.
- Promote collaboration and communication between marine operations and stakeholders to break down information silos.
- Establish KPIs and monitoring systems for contractor performance and vessel scheduling.

KPIs

- Percentage cargo in 14-day lookahead
- Payload utilization per voyage (%)

- On-time departure rate (%)

Lifting Equipment

Lifting equipment was underutilized, with few KPIs in place to monitor usage, allocation, or performance. The study recommended transitioning to an on-demand service model to maximize equipment use. Additionally, relevant KPIs should be developed to enable continuous monitoring and optimization of equipment deployment.

KPIs

- Crane & forklift utilization (%)
- Equipment downtime (hrs/month)

Vessel Management

There was no contractor performance management in place, and few KPIs were used to monitor the vessel fleet. Issues included vessel clustering at the quayside and a lack of oversight regarding fleet age, mix, and performance. The study recommended the adoption of an on-demand service model to reduce costs and enhance the efficiency of marine operations.

KPIs

- Cost per voyage (USD)
- Fleet fuel efficiency (tons cargo / ton fuel)

The figure below is a Summary of Findings

Operational Observations: Establishing a baseline

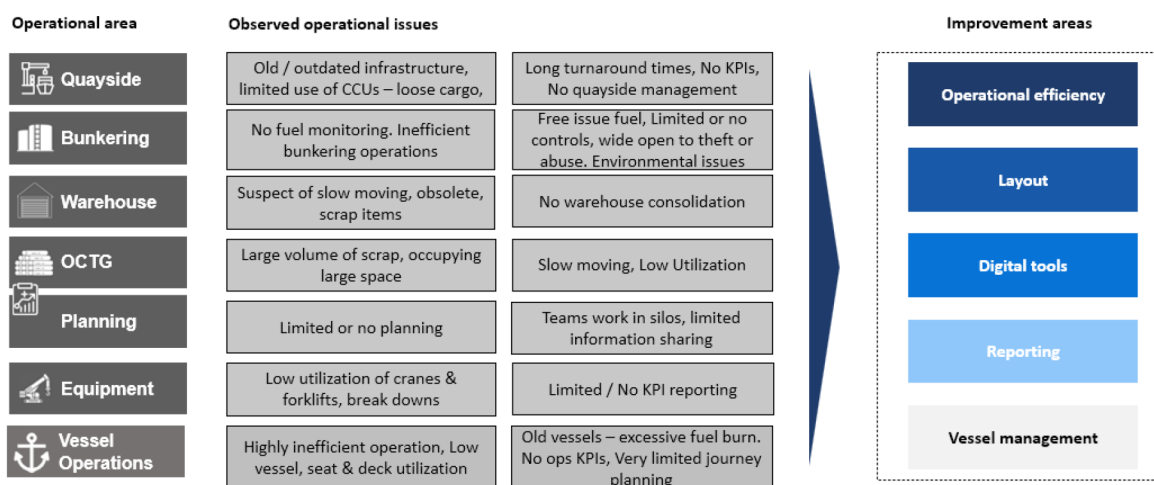


Figure 1—The output is consolidated to the figure above. This gives a concise output of 3 areas (a) Operational area; (b) Observed operational issues; (c) Improvement areas.

Multi-User Shore Base Facilities (MUSBFs): Concept and Benefits

Multi-User Shore Base Facilities are centralized hubs designed to serve multiple offshore operators. These facilities offer shared infrastructure such as quays, warehouses, offices, and fuel depots, alongside coordinated vessel and helicopter services. The concept draws from successful models in other industries, such as airport hubs and multi-tenant logistics parks, where shared infrastructure leads to economies of scale.

Key benefits include:

- **Cost Reduction:** Shared capital and operational expenditures across multiple users. Consolidation reduces duplication of infrastructure and personnel, leading to significant financial savings.
- **Efficiency Gains:** Optimized scheduling and resource utilization reduce vessel turnaround times. Integrated planning platforms enable precise coordination, minimizing idle time and maximizing throughput.
- **Environmental Benefits:** Consolidation reduces the number of trips, leading to lower greenhouse gas emissions and minimized ecological disturbance. Shared use of facilities can also enable investment in cleaner energy solutions, such as shore power and alternative fuels. This will be in line with the goals of the Blue Economy, a working methodology that allows for sustainability, digitalization, and the optimized use of infrastructure and resources.
- **Enhanced Collaboration:** Fosters partnerships between industry players, enhancing resilience and adaptability. Collaborative models encourage knowledge sharing, innovation, and best practices, contributing to industry-wide improvements.

Models of Shared Logistics

There are typically two operational models for shared logistics:

On-Demand Sharing: Resources are shared as needs arise, allowing flexibility but requiring robust real-time coordination. This model is akin to dynamic ride-sharing platforms in urban environments, optimizing available capacity as demand fluctuates.

Multiple users (operators, contractors) share logistics resources (e.g., vessels, helicopters, supply bases) only when they have overlapping or simultaneous needs. Logistics are triggered ad hoc based on actual demand. The key characteristics of such model include:

- Flexible and reactive: trips are organized when users require them.
- Users may "join" a trip initiated by another party if capacity is available.
- Coordination is dynamic, often relying on real-time communication.
- Efficiency depends on spontaneous alignment of needs.

Scheduled Sharing: Resources operate on fixed schedules, optimizing utilization and providing predictability. Scheduled sharing resembles public transportation systems, where regular, reliable services offer stability for planning and operations. The key characteristics of such model include:

- Predictable, timetable-based: "shared bus service" for logistics.
- Operators plan their supply and personnel movements around the fixed schedule.
- Often supported by formal agreements or contracts.
- Designed to maximize efficiency over time rather than short-term flexibility.

Empirical studies suggest that scheduled models often yield superior efficiency and sustainability outcomes. Scheduled models enable better forecasting, improved asset utilization, and stronger stakeholder alignment. However, hybrid models that combine scheduled and on-demand elements are increasingly being explored to balance flexibility and efficiency.

While not strictly a multi-user shore base case, Bourbon Offshore's partnership with Opsealog in the Gulf of Mexico region involved 25 platform supply vessels (PSVs) utilizing data-driven optimization. Their pilot achieved an average reduction of 45–50 tonnes CO₂ per vessel per month, improving both fuel efficiency and emissions footprint. This performance highlights the benefits of collaborative asset optimization and sharing operational data among stakeholders in shared logistical settings.¹

In Norway, the NSG E2E digital logistics platform—developed by Kongsberg Digital and NorSea Group—integrates warehousing, cargo handling, customs, and vessel scheduling at shared bases, delivering real-time coordination and visibility for multiple users Supply Chain Digital.²

Similarly, in the UK North Sea, the Central and Northern North Sea (CNNS) vessel-pooling scheme, facilitated by Peterson and the Aberdeen Marine Logistics Alliance, illustrates effective multi-user collaboration. ³Maersk Oil, Petrofac, and Dana Petroleum realized combined savings during 2014–2015 while cutting 800 tonnes of CO₂, equivalent to removing 275 cars from the roads annually⁴

These examples highlight the importance of stakeholder cooperation, robust governance structures, and investment in enabling technologies. Furthermore, they underscore the potential for Multi User Shore Base Facilities (MUSBFs) to drive broader sustainability agendas by incorporating renewable energy sources and energy-efficient infrastructure.

Challenges and Barriers to Adoption

Despite the clear benefits, several challenges hinder the widespread adoption of MUSBFs:

- **Cultural Resistance:** Operators accustomed to proprietary models may resist sharing, viewing collaboration as a threat to competitive advantage or operational autonomy. These could be an individual as well as organizational resistance. Studies show that on average about 24%⁵ of resistance is culturally based when a change is being adopted. Regular training and awareness sessions must be conducted to overcome this resistance.
- **Regulatory Complexities:** Varying national regulations can complicate shared operations. Issues related to customs, safety standards, and environmental compliance require harmonized frameworks to facilitate collaboration. The complexities could stem from multiple Authorities in variety of aspects for implementation. The change drivers, together with the stakeholders, should explain the advantages of the change that in turn shall reduce the regulatory complexities and might even widen the scope of MUSBFs.
- **Liability and Governance:** Clear frameworks are required to manage shared responsibilities, risks, and costs. Legal agreements must delineate roles, responsibilities, and dispute resolution mechanisms to ensure transparency and trust. Agreements must be framed in advance prior to execution of any operational activities from the trial phase itself. The lessons learnt approach during the trial phase shall ensure creating a robust business model.
- **Technological Integration:** Integrating diverse systems across multiple operators poses technical challenges. Standardizing data formats and platforms is essential to enable seamless coordination and real-time visibility. Diversified systems from numerous stakeholders pose a challenge for integration and from studies these can create disruptions mounting to about 60%⁵. Rigorous implementation strategies must be tried and tested prior to executing in real time Operations.

Operational Processes: Each NECs and IECs has its own operational process flows to be adhered to leading to a challenge in synchronizing these processes. Addressing these barriers requires a concerted effort from industry leaders, policymakers, and technology providers. Success hinges on building trust, aligning incentives, and demonstrating the tangible benefits of collaboration through pilot projects and early successes.

Strategic Implementation

For successful adoption, the following steps are recommended:

- **Stakeholder Engagement:** Early collaboration among operators, regulators, and service providers is critical. Engagement should focus on aligning objectives, identifying common interests, and

establishing shared visions for success together with utilizing the excess resources in more value-added activities.

- **Governance Structures:** Transparent, equitable agreements covering cost-sharing, liability, and access rights are fundamental. Governance models should balance flexibility with accountability, ensuring that all parties benefit fairly from shared operations.
- **Simplify Licensing:** Streamline the licensing and approval for operation of MUSBFs to encourage further interest and investment together with collaboration among existing operators and resources.
- **Digital Integration:** Leveraging digital twins, advanced planning tools, and integrated logistics platforms can optimize shared logistics. These technologies provide real-time visibility, predictive analytics, and scenario planning capabilities that enhance decision-making and efficiency.
- **Pilot Programs:** Starting with small-scale pilots can build trust and demonstrate value. Pilot projects offer opportunities to refine models, address challenges, and gather data to inform broader implementation.
- **Capacity Building:** Training and development initiatives can equip personnel with the skills and knowledge required to operate within shared logistics frameworks. Cultural change programs can also help shift mindsets towards collaboration and innovation.
- **Policy Support:** Governments and regulatory bodies can facilitate adoption by streamlining regulations, offering incentives for collaboration, and supporting infrastructure development through public-private partnerships.

Future Directions and Opportunities

The evolution of MUSBFs aligns with broader trends in the Blue Economy, including the integration of renewable energy, the transition to low-carbon fuels, and the adoption of circular economy principles. Future developments may include:

- **Integration with Renewable Energy:** MUSBFs can serve as hubs for offshore wind, tidal, and other renewable energy projects, supporting the logistics needs of these sectors while leveraging shared infrastructure. In 2024 approximately 12% of the overall renewable energy generations are from offshore facilities which is expected to grow to 18% by 2030⁶.
- **Decarbonization Initiatives:** Adoption of green technologies such as electric or hybrid vessels, shore power connections, and sustainable fuels can further reduce the environmental impact of logistics operations.
- **Digital Ecosystems:** Advanced data platforms can enable greater transparency, efficiency, and innovation within shared logistics networks. Integration with broader maritime and supply chain ecosystems will enhance value creation.
- **Resilience and Adaptability:** Shared facilities can enhance the resilience of offshore operations by providing flexible, scalable solutions that can adapt to changing market conditions and supply chain disruptions.

The Case for Multi-User Shore Bases and Shared Logistics in Offshore Oil & Gas in the Middle East

The Middle East has long been synonymous with large-scale, capital-intensive offshore oil and gas operations. Traditionally, these operations have been characterized by siloed logistical models, where each major operator maintains its own dedicated infrastructure, vessels, and support services. While this approach has historically served the needs of National Energy Companies (NECs) and International Energy Companies (IECs) in maintaining operational independence and security of supply, it increasingly conflicts with modern pressures for cost efficiency, sustainability, and resource optimization. In this context,

Multi-User Shore Bases (MUSB) and shared logistics models—whether on-demand or scheduled—present compelling opportunities for the offshore sector in the region.

In regions like the North Sea, such models have demonstrated substantial cost savings, improved asset utilization, and notable reductions in greenhouse gas emissions. For the Middle East, where offshore developments in Saudi Arabia, the UAE, and Qatar often operate in close geographic proximity, the logic of shared shore bases is equally applicable. Shared infrastructure would allow for the consolidation of supply runs, reduced duplication of assets, and streamlined regulatory compliance through centralized facilities. The accompanying map illustrates the geographic proximity of oil and gas fields located in three neighboring countries, underscoring their potential strategic advantage in adopting multi-user shore base (MUSB) facilities. Figure 2 overleaf shows a map of active and studied oil fields in the Persian Gulf.

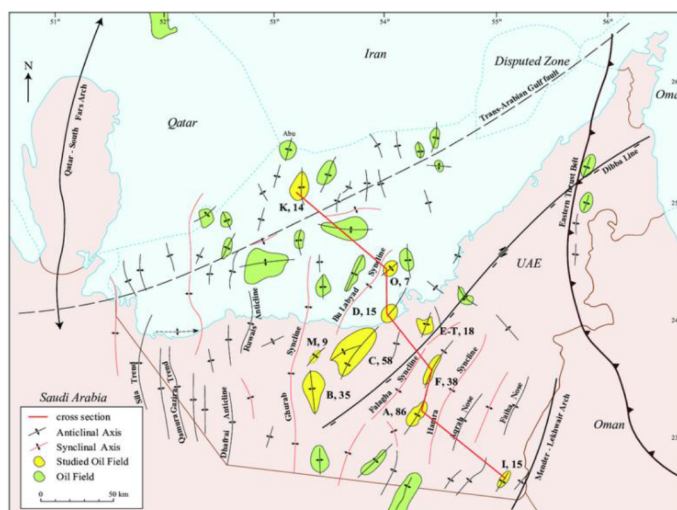


Figure 2—A map of active and studied oil fields.

Despite the benefits regulatory frameworks and procurement models often favor siloed arrangements. Nonetheless, the region's increasing focus on sustainability, driven by national visions such as Saudi Arabia's Vision 2030 and the UAE's Net Zero commitments and Vision 2050, is likely to accelerate interest in collaborative logistics models that align with decarbonization and efficiency goals.

Broader Implications for the Blue Economy

The adoption of MUSBFs not only benefits individual companies but contributes to the broader objectives of the Blue Economy by promoting sustainable marine resource use. Shared infrastructure reduces the environmental footprint of offshore operations, supports the transition to cleaner energy sources, and enhances the resilience of maritime supply chains. As global emphasis on sustainability intensifies, MUSBFs offer a practical mechanism for the offshore oil and gas sector to align with these objectives while maintaining competitiveness and operational excellence.

A fundamental principle of the Blue Economy is the optimization of ocean-based resources through multi-stakeholder collaboration, which aligns directly with the shared logistics approach that MUSBFs promote. By enabling collaboration between different offshore operators, service companies, and government bodies, MUSBFs facilitate the efficient use of existing marine infrastructure, reduce duplication, and promote economies of scale. These outcomes are central to the Blue Economy's goal of achieving long-term economic sustainability alongside environmental stewardship. The United Arab Emirates' accession to the Global Ocean Alliance in 2020 marked a significant step in reinforcing its commitment to the Blue Economy. This development contributed to the expansion of the Alliance's membership to a total of 72 nations, including key regional players such as Saudi Arabia, Qatar, Jordan, and Egypt.

Additionally, MUSBFs can play a significant role in accelerating the transition to greener technologies within the offshore sector. Shared logistics bases can serve as testing grounds and demonstration sites for emerging low-carbon technologies, such as electric-powered supply vessels, renewable energy integration for shore operations, and advanced waste management systems. These innovations not only enhance the environmental profile of offshore logistics but also provide scalable models that can be replicated across the wider maritime industry.

By concentrating logistics activities within centralized facilities, MUSBFs can stimulate economic development in coastal regions, supporting ancillary services such as maintenance, engineering, and hospitality. This clustering effect promotes knowledge exchange, innovation, and resilience within local communities, all of which are critical pillars of the Blue Economy.

MUSBFs also offer a strategic advantage in addressing global supply chain vulnerabilities. By consolidating logistics activities, these facilities enhance operational resilience against disruptions such as extreme weather events, pandemics, or geopolitical tensions. This resilience supports continuous offshore production, safeguarding energy security and contributing to the stability of global energy markets.

Finally, the data-driven nature of modern MUSBFs supports enhanced environmental monitoring and reporting, which are key components of the Blue Economy. By leveraging digital tools and platforms, shared logistics bases can provide accurate, real-time data on emissions, energy usage, and operational efficiency. This transparency not only helps operators meet regulatory requirements but also strengthens their social license to operate by demonstrating tangible commitments to sustainability and responsible resource management.

In summary, MUSBFs align closely with the foundational principles of the Blue Economy by promoting efficiency, collaboration, innovation, and sustainability. Their adoption has the potential to transform offshore marine logistics into a more resilient, environmentally responsible, and economically beneficial component of the global maritime sector.

Conclusion

Multi-User Shore Base Facilities (MUSBFs) are redefining offshore marine logistics by aligning operational efficiency with the Blue Economy's sustainability goals. By replacing siloed infrastructure with shared platforms, MUSBFs can reduce logistics costs by up to 30%⁷, while unlocking supply chain value and minimizing environmental footprints. This shift requires not only technology but also cultural and organizational change, underpinned by clear governance and stakeholder commitment.

Integrating digital technologies such as real-time analytics will be crucial in the successful management and operational control of MUSBFs. These tools improve transparency, optimize operations, mitigate risks, and boost efficiency, advancing both operational and ESG goals.

Beyond immediate gains, MUSBFs foster industry-wide collaboration, supporting shared innovation, better regulation, and sustainable infrastructure investment. They also aid de-risk emerging energy ventures like offshore wind, hydrogen, and carbon capture.

In a world facing constant volatility, geopolitical shifts, and energy transitions, MUSBFs offer the adaptability and resilience needed to maintain continuity and competitiveness.

In summary, Multi-User Shore Base Facilities align with the overarching principles of the Blue Economy, offering a viable solution to the inefficiencies of traditional siloed logistics. By embracing collaboration, leveraging digital innovation, and prioritizing sustainability, the offshore industry can enhance its competitiveness while contributing meaningfully to global environmental and socio-economic goals. The future of marine logistics lies in shared, integrated, and forward-looking approaches, such as having MUSBFs.

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